

Trauma in a Hemophiliac



Section I: Scenario Demographics

Scenario Title:	Pediatric Trauma in a Hemophiliac		
Date of Development:	25/04/2016 (DD/MM/YYYY)		
Target Learning Group:	☐ Juniors (PGY 1 - 2)	☒ Seniors (PGY ≥ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Donika Orlich
Affiliations/Institution(s):	McMaster University
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Section III: Curriculum Integration

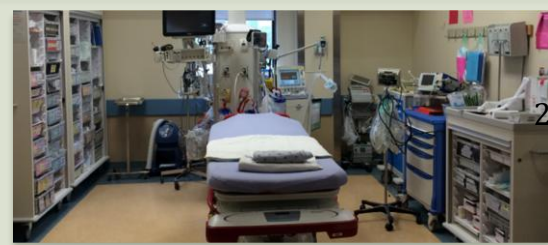
Learning Goals & Objectives	
Educational Goal:	To expose learners to a pediatric trauma in a patient with Hemophilia
CRM Objectives:	1) Demonstrates leadership by maintaining control in a crisis and making prompt and firm decisions 2) Demonstrates an organized and efficient problem-solving approach and collaborates with team while doing so
Medical Objectives:	1) Initiates factor replacement for bleeding in the hemophiliac patient 2) Performs a capacity assessment and outlines a plan of action for the incompetent patient

Case Summary: Brief Summary of Case Progression and Major Events
A 16-year-old female presents following an MVC. Past medical history is significant for hemophilia A. She has a laceration on her arm and a bruise on her forehead, but denies HA/N/V. The learner should recognize high potential for bleeding, and implement immediate treatment with rVIII replacement, along with pan-CT imaging. The CT head will show a small ICH. The patient wants to leave AMA following normal CT results, and the learner must perform a capacity assessment and outline a plan of action for the incompetent patient. The patient should be sedated and/or intubated anticipating decline using neuroprotective measures. Consults should be made to the ICU and hematology.

References
Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). <i>Rosen's emergency medicine: Concepts and clinical practice</i> . St. Louis: Mosby.
Website: E-medicine (accessed April 25, 2016) http://emedicine.medscape.com/article/779322-treatment#d9



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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☐ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
		<i>Select most important dimension(s)</i>	
Confederates			
Brief Description of Role			
Nurse	To assist with medication administration, clarify doses from learners, and prompt as to the status of the patient.		
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☒ Other: rVIII replacement	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
Laceration to right arm (5cm), bleeding actively. Bruise over R temporal area			
E. Approximate Timing			
Set-Up:	5 min	Scenario:	20 min
		Debriefing:	40 min

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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

You are working in a level three trauma center and are told that EMS just arrived from an MVC involving a 16-year-old female passenger who has known hemophilia. Vitals are stable. She has a laceration to her arm, and a bruise on her head, but has GCS 15 and only complains of arm pain.

B. Patient Profile and History

Patient Name: Cintia Puzenski	Age: 16	Weight: 50kg
Gender: ☐ M ☒ F	Code Status: Full	

Chief Complaint: Right arm pain.

History of Presenting Illness: MVC about 50km/hr. She was belted passenger in front seat. No injuries to driver (mother). No airbags deployed.

Past Medical History:	Hemophilia A Previous admissions for spontaneous joint bleeds x3 and spontaneous epistaxis x1, all requiring transfusion	Medications:	No prior humate P or DDAVP rVIII prophylaxis 30IU/kg IM M/W/F
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Allergies: None

Social History: Attends high school. Lives with parents. No EtOH, drugs, smoking

Family History: No family history of hemophilia, other bleeding disorders

Review of Systems:	CNS:	Denies HA, N/V. No vision changes. No paresthesias.		
	HEENT:	Denies neck pain.		
	CVS:	No CP, no presyncope		
	RESP:	No SOB. No hemoptysis		
	GI:	No abdominal pain.		
	GU:	Hasn't voided since accident.		
	MSK:	Pain R arm (still bleeding)	INT:	Nil

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display		☒ Monitor On, no data displayed		☐ Monitor on Standard Display	
HR: 110/min		BP: 120/80		RR: 14/min	
Rhythm: sinus		T: 36°C		Glucose: 5 mmol/L	
				O ₂ SAT: 100% RA	
				GCS: 15 (E4 V5 M6)	
General Status: Alert & Oriented. Complaining of R arm pain/bleeding.					
CNS:	Alert and oriented with GCS 15. No focal neuro deficits. C-collar in situ.				
HEENT:	PERRL 4mm. No signs basilar skull #. Bruise R temporal area (non-boggy). Airway clear.				
CVS:	Normal S1, S2. No murmurs.				
RESP:	GAEB. No crackles. No crepitus.				
ABDO:	Soft. No bruising. Non-tender.				
GU:	No hematuria with foley.				
MSK:	5cm superficial lac R forearm with active bleeding			SKIN:	Normal colour

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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: sinus HR: 110/min BP: 120/80 RR: 14/min O ₂ SAT: 100% T: 36°C	Comfortable but complaining of R arm pain. ++ bleeding from arm laceration	<u>Learner Actions</u> ☐ IV, monitors ☐ Labs (CBC, INR, G&S, factor VIII level, BhCG, basic labs) ☐ 1L IV NS bolus ☐ ABCDE primary survey ☐ Control local bleeding ☐ Analgesia ☐ rVIII 50U/kg ☐ TxA 1g IV over 10min ☐ FAST US exam ☐ Portable CXR, PXR ☐ CT Head, C-spine, Chest, Abdo, Pelvis ☐ Arm x-ray ☐ Hematology consult	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - Analgesia given → HR 90 - Laceration sutured/pressure dressing applied → bleeding slows - Laceration not addressed by 3 minutes → nurse to cue “she’s bleeding a lot from that arm” <u>Triggers</u> <i>For progression to next state</i> - 8 minutes or all learner actions complete → 2. Back from CT (assume 20 minutes passed)
2. Back from CT Rhythm: sinus HR: 95/min BP: 125/80 RR: 12/min O ₂ SAT: 100%	**At state onset, verbal report from radiology “small ICH, no shift” Patient initially complains of mild HA. Vomit x1 in scanner. Two minutes into state, patient wants to leave AMA	<u>Learner Actions</u> ☐ Re-assess patient ☐ rVIII & TxA if not already given ☐ Check cap sugar ☐ IV/IM sedation for agitation ☐ ± Plan for intubation ☐ Neurosurgery consult ☐ Hematology consult (if not yet done)	<u>Modifiers</u> - No sedation by 12 mins → patient pulls out IV & becomes verbally aggressive <u>Triggers</u> - Verbalize plan for intubation → 3. Intubation - No plan for intubation but patient sedated and actions complete → 4. Labs Back
3. Intubation Rhythm: sinus HR: 100/min BP: 120/80 RR: 15/min O ₂ SAT: 100 %	Patient calm after sedation given	<u>Learner Actions</u> ☐ RA Pupils/neuro exam prior to paralysis ☐ Pre-oxygenate with 100% ☐ Neuro-protective measures ☐ Elevate head of bed ☐ Post-intubation CXR & sedation	<u>Modifiers</u> - Propofol given → BP 95/70, HR 105 <u>Triggers</u> - Intubation complete → 4. Labs Back

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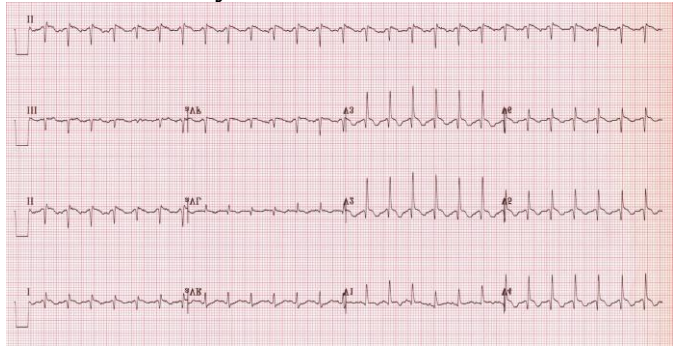
4. Labs Back Rhythm: sinus HR: 85/min BP: 115/75 RR: 12/min O ₂ SAT: 100%	* start state by giving labs to team	<u>Learner Actions</u> - ☐ ICU consult - ☐ Discuss with Heme if any further management required	<u>END CASE PRN</u>
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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results						
Na: 140	K: 4	Cl: 105	HCO ₃ : 25	BUN: 5	Cr: 80	Glu: 5.5
VBG	pH: 7.38	PCO ₂ : 40	PO ₂ : 95	HCO ₃ : 25	Lactate: 2.2	
WBC: 5		Hg: 110		Hct: 0.4		Plt: 250
INR: 1.0		PTT: 45 (increased)				

Images (ECGs, CXRs, etc.)	
Normal CXR  CXR source: https://radiopaedia.org/cases/normal-chest-radiograph-female-1	Normal Pelvis X-ray  PXR source: http://radiopaedia.org/articles/pelvis-1
Normal R Forearm  Xray source: http://www.auntminnie.com/index.aspx?sec=ser&sub=def&pag=dis&ItemID=56736	ECG: sinus tachycardia  ECG source: https://lifeinthefastlane.com/ecg-library/sinus-tachycardia/
Ultrasound Video Files (if applicable)	
No FF on FAST	No pericardial fluid

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) How did you tailor your approach to trauma in this hemophiliac patient? 2) What was your approach to managing this patient wanting to leave AMA? 3) What would have been your ultimate disposition for this patient had the CT head been normal? 4) Did you feel comfortable with the management of this patient? What are the key elements to take away from a patient like this to increase comfort in the future? How else would you access this information?	
Key Moments	
1) Recognize head injury as a carrying severe bleeding risk and initiating prompt rVIII replacement immediately (even in the absence of symptoms)	
2) Makes early consultation with hematology	
3) Capacity assessment	